3.2 The garden in the margin

Everything in the left margin of this page is drawn by TEX at shipout time from a few hundred lines of TikZ. There are no image files, no imported paths and no external programs; every petal is a parametric curve evaluated while the page is being written. This section records how it is done, so that the margin documents itself.

Definition 3.1. A *primitive* is a closed outline $\gamma\colon [0,1] \to \mathbb{R}^2$ authored by hand together with a small set of free parameters $(p_1, \dots, p_k)$. An *instance* of the primitive samples the parameters from fixed intervals and draws γ in a local frame chosen by the caller. The composition is *authored*; only the parameters are *sampled*. This is what "procedural" means here: no flower is grown from a grammar, but no two petals share a shape.

Construction 3.2 (Petal)**.** A petal of length 1 pointing along $+y$ is the egg curve

$$\begin{aligned} x(\theta) &= w \cos\theta \,(1 + 0.35 \sin\theta)\,\big(1 + \rho(\theta)\big), \\ y(\theta) &= \tfrac{1}{2} + \tfrac{1}{2}\sin\theta\,\big(1 + \rho(\theta)\big), \qquad \theta \in [-90^\circ, 270^\circ), \end{aligned}$$

with half-width $w \in [0.46, 0.62]$ and a *ruffle*

$$\rho(\theta) = a\,\frac{1+\sin\theta}{2}\,\big(\sin(5\theta + \varphi_1) + \tfrac{1}{2}\sin(11\theta + \varphi_2) + \tfrac{1}{4}\sin(23\theta + \varphi_3)\big),$$
$$a \in [0.028, 0.046].$$

The factor $(1+\sin\theta)/2$ vanishes at the base and is largest at the tip, so the petal attaches cleanly and frays where petals fray. The phases φ_k are drawn once per petal; after that ρ is deterministic, which matters because the same outline is needed three times: once as a gradient fill, once as a clipping path and once as a hairline stroke. The fill is an axis gradient from `ink!22` at the base to `ink!95!black` at the tip. Inside the clip, nine dark and six pale lens-shaped streaks radiate from the base at random angles, seven hairline veins run to the edge, and the outline is stroked again 1.3 pt wide so that only its inner half shows, giving the dark rim. The brushed texture of the petal is nothing more than those fifteen translucent lenses.

Construction 3.3 (Flower)**.** Five petals are placed at 72° intervals with ±7° of jitter and lengths in $[0.83, 1.03]$; the whole is rotated by a random angle. A white disc with a fuzzy-edged *fading* is laid over the centre, which is what pulls the petal bases toward white without any seam. Sixteen stamens of random length end in gold dots; seven more gold dots sit at the heart.

Construction 3.4 (Leaves)**.** A leaf of length L along $+x$ has upper and lower edges

$$\begin{aligned} x(t) &= L\,\frac{1-\cos t}{2}, \\ y_{\pm}(t) &= \pm\,\lambda L \sin t\,\big(1 + s\,(1 - |\sin(nt + \psi)|)\big) + c\,L\Big(\tfrac{1-\cos t}{2}\Big)^2, \end{aligned}$$

for $t \in [0°, 180°]$. The width ratio λ, serration depth s, tooth count n and curl c distinguish the species: $(\lambda, s, n) = (0.26, 0.10, 6)$ gives a serrated leaflet, $(0.08, 0, 1)$ with $c \in [-0.35, 0.35]$ a grass blade. The two edges are separate smooth plots joined at the tip, so the tip is pointed. A *dissected* leaf is a rib carrying three pairs of serrated leaflets of decreasing size at $\pm 52°$ and a terminal leaflet; a *fern* is a rib with leaflets placed by path markings and scaled by their sequence number so that they taper.

Construction 3.5 (Sprays and umbels)**.** A *spray* is a ternary recursion of depth d: each segment forks into three segments at $-40°, 0°, +40°$ ($\pm 14°$) of 0.6 its length ($\pm 12\%$), and at depth 0 draws a five-petal floret two millimetres across and a couple of stray dots. An *arch spray* is one long bent stalk with depth-2 sprays hung alternately left and right along it. An *umbel* fans eleven rays upward, each ending in a secondary umbel of six short rays with a dot at every tip; four short bracts curl up from the base. Tendrils are logarithmic spirals $r = 0.06\, e^{0.0036\,\theta}$.

3.2.1 Composition

Two stalks run the height of the page: a thick vine and a thinner companion that crosses it four times. Each is a smooth plot through eleven control points whose x coordinates carry ± 3 mm of jitter. Leaves are not placed by hand; they are hung along the stalks by `decorations.markings`, which executes arbitrary drawing code at fractions of the arc length in a frame aligned with the tangent, so every leaf follows the direction of its stem. The same mechanism drops named coordinates at six fractions of the vine, and the larger elements — flowers, bells, dissected leaves, the arch sprays — are attached to short side branches grown from those anchors. Elements are drawn in depth order: sprays and ferns first, foliage, then bells and buds, then flowers, then the scattered ornaments. Everything is clipped to the margin column.

Remark 3.6 (Procedural is not random)**.** The seed of the generator is set to a function of the page number before the garden is drawn. Page n therefore receives the same garden on every compilation, and a different one from page $n+1$. The variation the reader notices between pages is real; the reproducibility is what makes the document stable enough to typeset.

Proposition 3.7. *Let P be the number of primitive instances on a page and S the mean number of samples per outline. The cost of a page is $\Theta(PS)$ evaluations of `pgfmath`, and in practice on the order of a minute per page for $P \approx 400$ and $S \approx 50$ under `pdflatex`.*

Proof sketch. Each outline is evaluated once for its fill, and its clip and stroke reuse the saved soft path (`save path`/`use path`), so the per-primitive cost is the plot itself; the dozen or so textural strokes inside the clip are constant work. The constant is dominated by `pgfmath`, which computes in TEX dimensions: integers and intermediate results must stay below 16 384, which is also why the page seed is $97\,n + 1030$ and not something larger. □



Exercise 3.8. The composition above is authored. Replace it by a growth process in the spirit of Wong, Zongker and Salesin (*Computer-generated floral ornament*, SIGGRAPH 1998): starting from one seed on the vine, repeatedly choose a primitive, find the largest free disc in the margin region that touches an existing element, and place the primitive there if it fits, until no disc larger than a threshold remains. The ornament then adapts to any text block, and a page with a wide display equation grows a wider garden. Pure `pgfmath` is a poor host for the bookkeeping; `luatex` with the layout computed in Lua and the drawing left to the primitives above is the natural implementation.

3.2.2 The growth stage

This document is the answer to the exercise. Its margin is not the authored composition of the previous section: nothing in it was placed by hand. A small Lua program, run by `luatex` inside the shipout hook, decides where every element goes and how large it is, and then calls the same primitives to draw them.

Definition 3.9. A *proxy* for a motif is a disc of radius $r(s)$, where s is the motif's scale, that approximates the area it occupies. A point p of the free region R has *free radius*

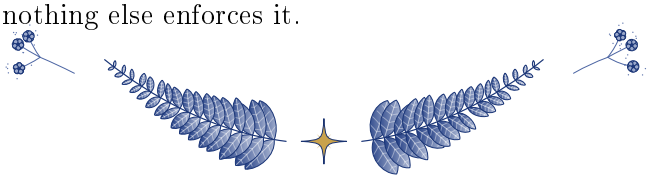
$$f(p) = \min\Big(d(p, \partial R),\ \min_i \big(\|p - c_i\| - \kappa\, r_i \big) \Big),$$

the radius of the largest disc about p that stays inside R and overlaps no existing proxy (c_i, r_i) by more than the fraction $1 - \kappa$; here $\kappa = 0.82$.

Construction 3.10 (Growth)**.** The region R is the margin column, and on the last page also the rectangle between the end of the text and the footer, if that is at least a few centimetres tall. R is seeded with a vine (and, when the second rectangle exists, a branch that leaves the vine and waves across it); the vine's samples are the initial *anchors*. Then, until the largest free radius falls below the smallest motif:

1. on a grid of candidates spaced 2 mm apart, take the point p that maximises $f(p)$ among those within 2.1 cm of an anchor, with a random factor in $[0.85, 1.15]$ so that ties break differently on every seed;
2. choose a motif whose smallest proxy fits $f(p)$, by weight, respecting a cap per page on each kind; then set its scale so that $r(s) \approx 0.97\, f(p)$;
3. orient it away from the nearest anchor, draw a stem from that anchor to its base, and add its proxy to the set and its centre and base to the anchors.

Because the largest hole is always taken first, flowers and dissected leaves are placed early, sprays and ferns fill the middle, and the loop ends by tucking buds, tendrils and small ornaments into whatever gaps remain. That ordering is the whole reason the result looks *composed* rather than scattered; nothing else enforces it.



Remark 3.11 (Adapting to the text)**.** The free region is computed from the page layout, so changing the margins moves the garden; and the end of the text is recorded with `zref-savepos`, so on a short last page the garden spills into the space below the words. The page you are reading ends before the bottom of the text block, which is how the spill below came to be. One extra compilation is needed for the recorded position to take effect.

Remark 3.12 (Cost)**.** The layout itself is cheap: a few hundred candidates times a few hundred placements in Lua takes well under a second. The drawing costs what it did before, since the primitives are unchanged and `pgfmath` is still doing the curve evaluation in TEX.

3.2.3 Building and sharing

The package is one file, `florilegium.sty`, plus `grow.lua` for the grown composition; the demonstration documents are a short preamble and three text files. The authored document compiles with `pdflatex` in about half a minute per page, and with `lualatex` or `xelatex` unchanged; the grown one needs `lualatex`, run twice. To share it, the three routes that work are, in increasing order of effort: paste the files into a blank Overleaf project and share the link; put them in a Git repository with a continuous integration job that compiles the PDFs on every push and attaches them to tagged releases; or publish the package so that any document can `\usepackage` the garden and choose its own seed, column and hue.

Remark 3.13. Two mistakes cost an afternoon each and are worth recording. A `\foreach` variable must not be an existing control sequence; `\fi` in particular is a TEX primitive and silently destroys the conditional that contains it. And a coordinate whose component contains parentheses, such as $(\sin\alpha, \cos\alpha)$, must have each component wrapped in braces, because TikZ reads a coordinate up to the first closing parenthesis it meets.